

Afterburners

CLASS 7

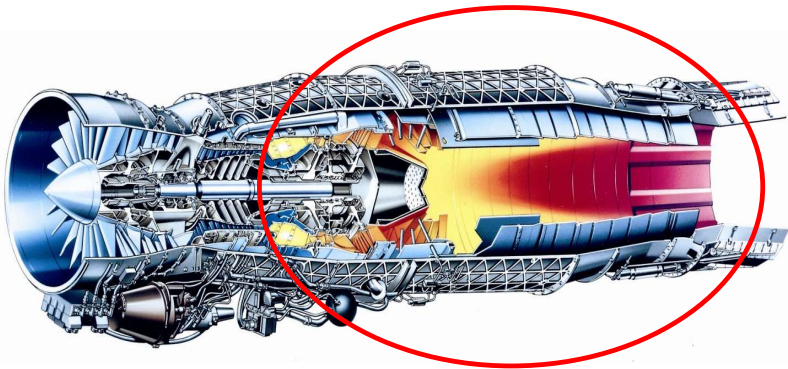


Introduction

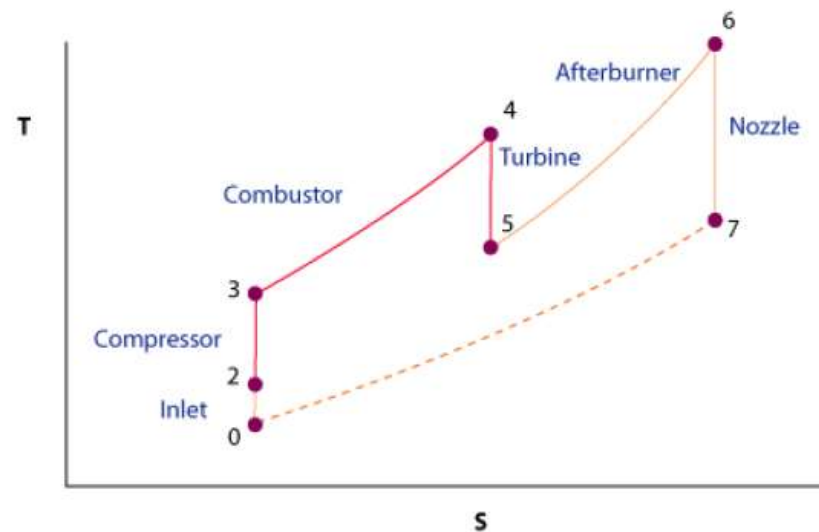
Introduction to the conceptual design and optimization of afterburners based on the fundamentals of combustion and fluid dynamics.

Concept

Afterburning or reheating is one method of augmenting (increasing) the basic thrust of the aircraft engine, when required, without having to use a larger engine with its concurrent penalties of increased area and weight.



Turbojet-Afterburner Cycle



Requirement

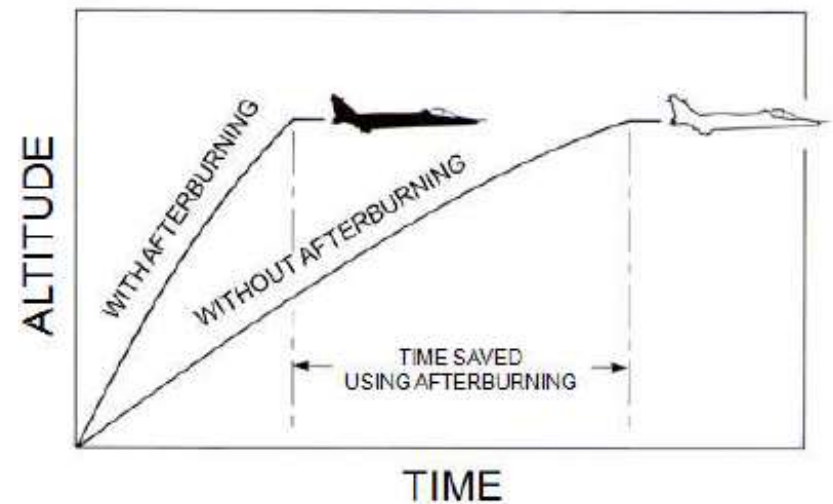
What is the driving requirement for afterburners?

Military Aviation

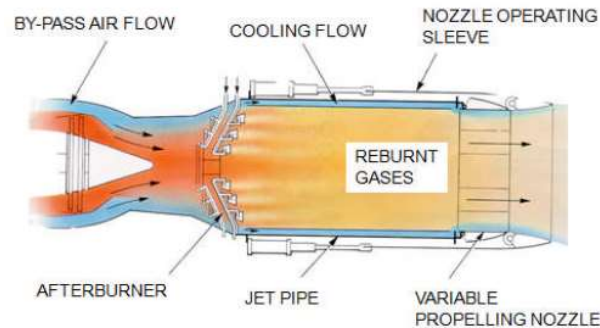
- Take off (short runways, fast take off) and high supersonic speed (combat).

Supersonic civil aviation

- Takeoff and acceleration above sonic speeds



Thus, the thrust of any given engine may be increased by raising the temperature of the exhaust gases. Fortunately, this is easy to do, since less than one-third of the total oxygen in the air flowing through the engine is consumed in the main combustion chamber, and the remainder is available in the turbine efflux gases to support further combustion. Thus, in order to obtain more thrust, it is only necessary to inject additional fuel into these gases to mix and burn with the available oxygen.



How it Works?

$$F_N = \dot{m}V + A_N(p_N - p_A)$$

$$F_N = \dot{m}V$$

$$V_E \propto T_E^{\frac{1}{2}}$$

$$\frac{V_R}{V_E} = \left(\frac{T_R}{T_E}\right)^{0.5}$$

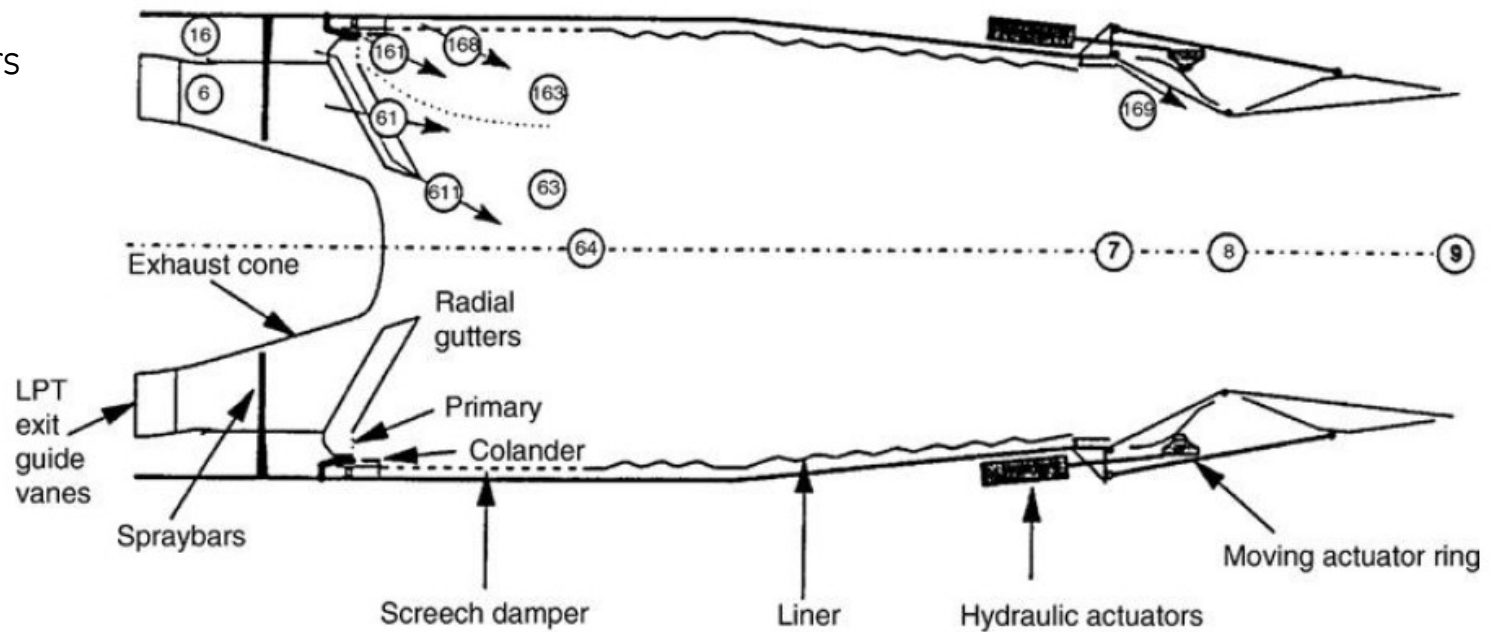
- Assumptions:
 - Additional fuel flow and pressure loss neglected

Main Differences

The afterburner is not subjected to the physical and temperature limits of a combustor as the outlet of the afterburner passes through a fixed element (nozzle) rather than a rotating element (turbine) as in the case of combustion chamber. The temperature rise is limited by the amount of oxygen that is available for combustion and limited by the liner and nozzle cooling air.

General Arrangement of Conventional Turbojet Afterburner

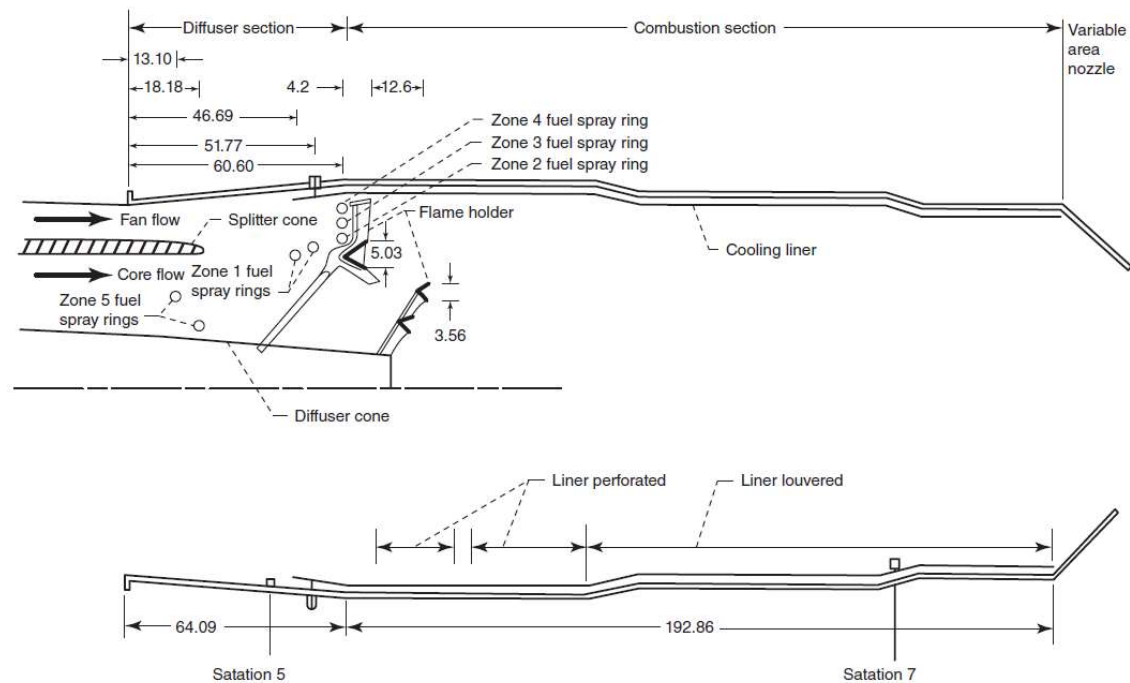
- Diffuser
- Flameholders
- Liner
- Nozzle



EJ200 afterburner and nozzle

General Arrangement of Conventional Turbofan Afterburner

- Diffuser
- Flameholders
- Liner
- Nozzle



Afterburner for TF30-P3 augmented turbofan engine [all dimensions are in centimeters]

Design Criteria

Afterburners Design Criteria



LARGE
TEMPERATURE
RISE



LOW DRY LOSS



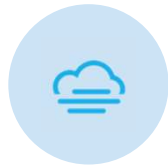
WIDE
TEMPERATURE
MODULATION



HIGH
COMBUSTION
EFFICIENCY



SHORT LENGTH;
LIGH WEIGHT



ALTITUDE LIGHT –
OFF CAPABILITY



NO ACOUSTIC
COMBUSTION
INSTABILITIES



LONG LIFE, LOW
COST, EASY REPAIR

Afterburners Design Criteria: Temperature Rise

Nozzle exit temperature $\approx 1850 - 2100$ K

Limitations

- Dissociation
 - Is an endothermic reaction which removes heat from the system
- Cooling Requirements:
 - About 10% of the turbine delivery air is needed for afterburner wall cooling, air will mix downstream of the afterburner lowering the average exit temperature.
- Vitiation
 - Temperature rise limited because of oxygen depletion in the main burner
- Reheat Buzz
 - Noise generated by combustion associated with rich extinction of flame (increase in FAR, length, low pressure), prevalent buzz may result in mechanical damage.

Afterburners Design Criteria: Temperature Rise

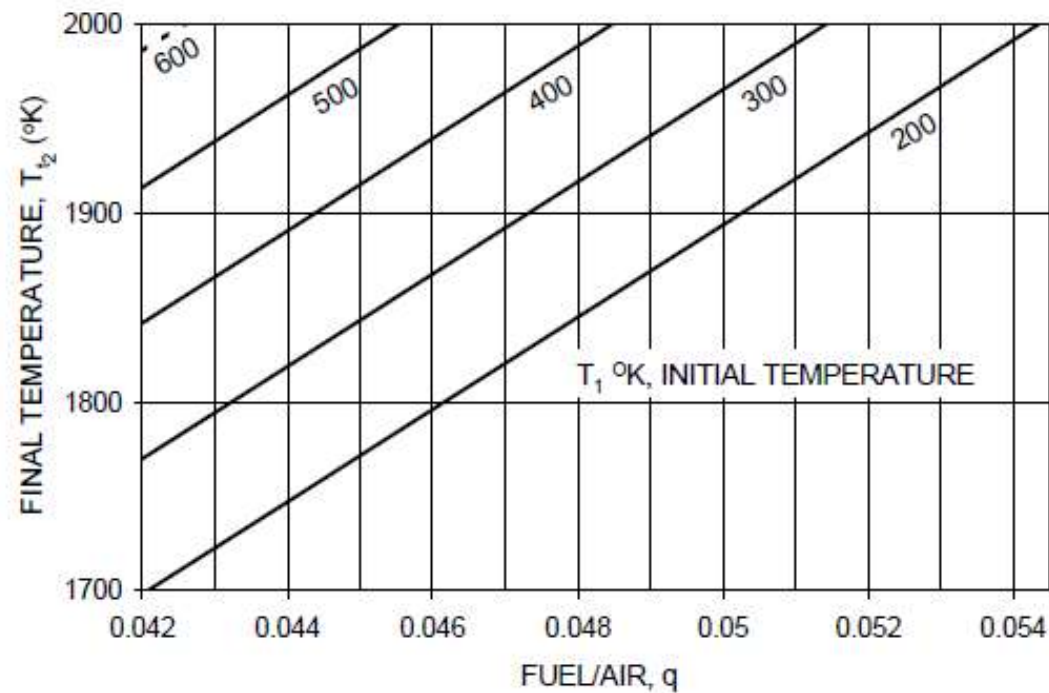
Temperature rise can be calculated either:

- By solving energy balance equations (different levels of accuracy):

$$\eta_{AB} = \frac{(\dot{m}_0 + \dot{m}_f + \dot{m}_{f,AB})h_{AB_{exit}} - (\dot{m}_0 + \dot{m}_f)h_{turbine_{exit}}}{\dot{m}_{f,AB}FHV}$$

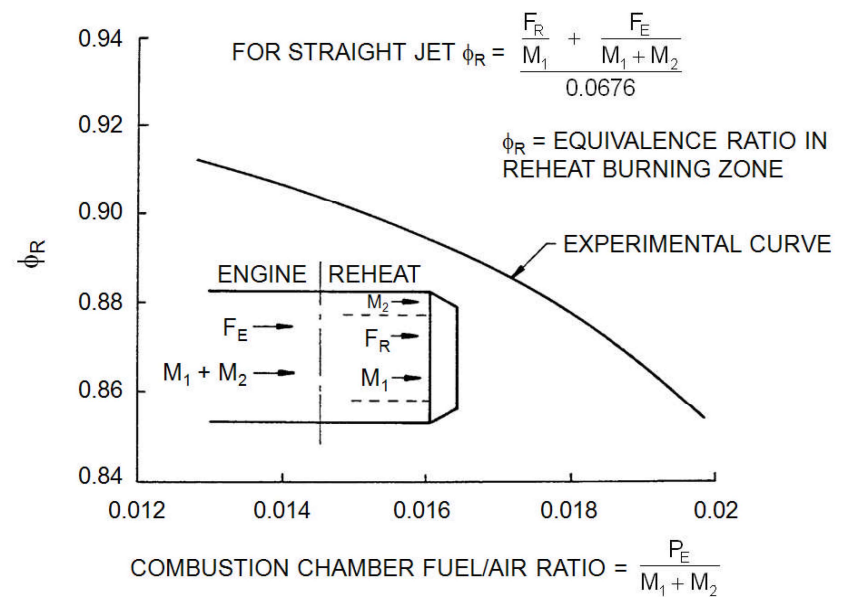
- Using afterburner temperature rise characteristics (only for first order approximations as they do not account for dissociation)

Afterburners Design Criteria: Temperature Rise Characteristics

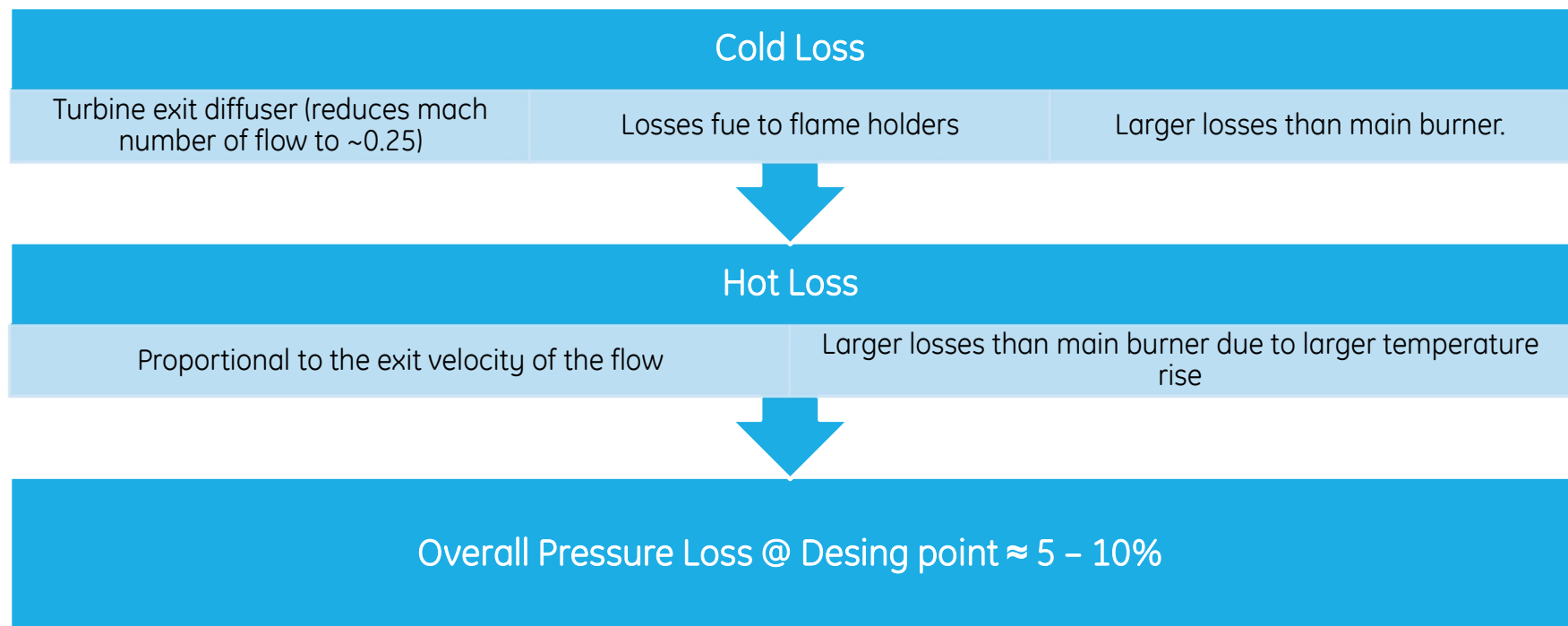


Afterburners Design Criteria: Vitiation Factor

Is a direct relation between the engine real fuel to air ratio and the oxygen depletion in the reheat section



Afterburner Design Criteria: Losses



Afterburners Components

Afterburners Components: Diffuser

Function: Reduces the flow velocity in the jet pipe ($M \approx 0.2$).

- Reduces cold pressure loss associated with flame holders and fuel injectors
- Classical diffuser theory
 - Optimum angle $\approx 7^\circ$
 - Real $\approx 11^\circ$ shorter diffuser (flame holders reduce the separation loss associated with high cone angles)

Afterburners Components: Flame holders

Commonly designed as a “<” shape (“V” Shaped gutters)

- Compromise between minimum drag, thermal integrity, stability and flame propagation

In high temperature streams (>800K):

- Enough heat release to sustain a pilot zone and combustion
- Flame spreading rate = $f(\text{turbulent flame speed and gas velocity})$

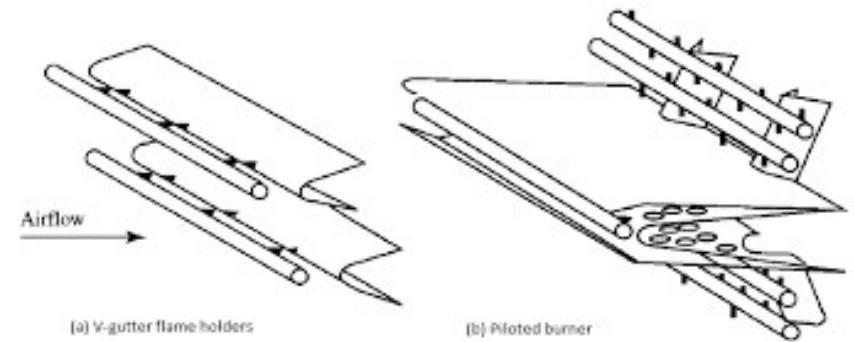
In “lower” temperature streams:

- Necessary to separately meter fuel and air into a pilot zone to ensure sufficient heat release for flame propagation.

Afterburner fuel is not uniformly distributed but confined to central section

- Allows close to stoichiometric combustion
- Combustion zone is surrounded by “relatively” cool gas preventing overheating of jet pipe walls

Afterburners Components: Flame holders



Afterburners Components: Flame holders

The process of flame stabilization is achieved via flameholders, that creates massively separated wakes and is responsible for the bulk of the total pressure loss.

Static pressure ratio is expressed in terms of the inlet and exit Mach numbers and the flameholder drag coefficient C_D as

$$\frac{p_e}{p_i} = \frac{1 + \gamma_i M_i^2 \left(1 - \frac{C_D}{2}\right)}{1 + \gamma_e M_e^2}$$

where C_D is expressed as

$$C_D = \frac{D_{flameholder}}{q_i \cdot A_{flameholder}}$$

Afterburners Components: Flame holders

Flameholder drag coefficient also influences the duct area in the afterburners and therefore the exit Mach number at the flameholders.

$$A = \left(\frac{1 + \gamma_i M_i^2 \left(1 - \frac{C_D}{2} \right)}{\gamma_i M_i} \right) \sqrt{\frac{\gamma_i - 1}{1 + \frac{\gamma_i - 1}{2} M_i^2}}$$

And

$$M_e^2 = \frac{2(\gamma_e - 1) - \gamma_e A^2 + \sqrt{(\gamma_e A^2 - 2\gamma_e + 2)^2 + 2(\gamma_e - 1)^2(A^2 - 2)}}{(\gamma_e - 1)(A^2 - 2)}$$

Afterburners Components: Flame holders

With exit Mach number known, we may use isentropic relationships and the static pressure ratios to calculate the total pressure ratio

$$\frac{P_e}{P_i} = \frac{p_e}{p_i} \frac{\left(1 + \frac{\gamma_e - 1}{2} M_e^2\right)^{\frac{\gamma_e}{\gamma_e - 1}}}{\left(1 + \frac{\gamma_i - 1}{2} M_i^2\right)^{\frac{\gamma_i}{\gamma_i - 1}}} = \frac{1 + \gamma_i M_i^2 \left(1 - \frac{C_D}{2}\right)}{1 + \gamma_e M_e^2} \frac{\left(1 + \frac{\gamma_e - 1}{2} M_e^2\right)^{\frac{\gamma_e}{\gamma_e - 1}}}{\left(1 + \frac{\gamma_i - 1}{2} M_i^2\right)^{\frac{\gamma_i}{\gamma_i - 1}}}$$

Afterburners Components: Flame holders

The analysis of an afterburner in the “wet” mode differs from the “dry” mode only in the energy balance mentioned before. By neglecting the fuel to air ratio in the afterburner in favor of 1 on the left-hand side of the energy equation and renaming the product term on the RHS as “q”, results in

$$\frac{C_{pe}T_e}{C_{pi}T_i} \cong 1 + \frac{q}{C_{pi}T_i}$$

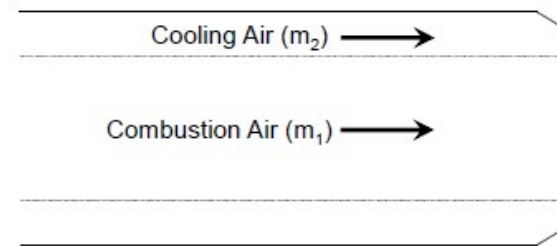
Therefore

$$A = \left(\frac{1 + \gamma_i M_i^2 \left(1 - \frac{C_D}{2} \right)}{\gamma_i M_i^2} \right) \sqrt{ \frac{\gamma_i - 1}{1 + \frac{\gamma_i - 1}{2} M_i^2} \left(\frac{1}{1 + \frac{q}{C_{pi}T_i}} \right) }$$

Basic Design features: Jet Pipe Cooling

The amount of jet pipe cooling that is required will depend on the engine cycle and on the amount of reheat thrust boost the system has been designed to achieve.

A design decision is taken to ensure that the flame spreading nearest to the jet pipe is such that no additional jet pipe wall cooling is required. This is in effect equivalent to setting up a large film-cooling zone on the jet pipe inner wall to ensure convective heating from inlet to the nozzle exit. The reduced thrust boost is offset against a reduced weight and complexity



F_E = main combustor fuel flow rate (kg/s)

F_R = afterburner fuel flow rate (kg/s)

m_1 = air used in afterburner combustion (kg/s)

m_2 = air used for afterburner cooling (kg/s)

q_{ov} = FAR required to raise air from T_1 to T_R

q_E = FAR required to raise air from T_1 to T_4 (main combustor FAR)

T_1 = engine inlet total temperature (K)

T_4 = main combustor outlet total temperature (K)

T_R = afterburner outlet total temperature (K)

Afterburners Components: Liner

Liner size diameter is a trade off between nozzle weight and pressure loss

- Nozzle weight $\sim D^3$
- Pressure loss $\sim \frac{1}{D^4}$
- Depends on engine application
 - If the application calls for long periods of afterburning the performance requirement will probably outweigh that of size and weight.

$$\eta_c = f \left[\frac{P^{1.4} \cdot A \cdot L \cdot e^{\frac{T}{1000}}}{m} \right]$$

Afterburners Components: Liner Cooling

$$f_{ov} = \frac{\text{total fuel}}{\text{total air}} = \frac{\dot{m}_{fuel} + \dot{m}_{AB}}{\dot{m}_1 + \dot{m}_2} = \frac{\dot{m}_{fuel}}{\dot{m}_1 + \dot{m}_2} + \frac{\dot{m}_{AB}}{\dot{m}_1 + \dot{m}_2}$$

$$f_{ov} = f_{GG} + \frac{\dot{m}_{AB}}{\dot{m}_1 + \dot{m}_2} \text{ or } \frac{\dot{m}_{AB}}{\dot{m}_1 + \dot{m}_2} = f_{ov} - f_{GG}$$

$$\frac{\dot{m}_{fuel}}{\dot{m}_1 + \dot{m}_2} + \frac{\dot{m}_{AB}}{\dot{m}_1} = 0.0676 \text{ (for stoichiometric combustion)}$$

$$\frac{\dot{m}_1}{\dot{m}_1 + \dot{m}_2} = \frac{f_{ov} - f_{GG}}{0.0676 - f_{GG}} \text{ (} f_{ov} \text{ \& } f_{gg} \text{ estimated from charts)}$$

$$\frac{\dot{m}_1}{\dot{m}_1 + \dot{m}_2} = \frac{f_{ov} - f_{GG}}{\eta_b(\phi_{AB} \cdot 0.0676 - f_{GG})} = \frac{\text{diameter of combustion zone}}{\text{diameter of reheat pipe}}$$

ϕ_{Ab} = vitiation factor

η_b = burner efficiency

Afterburners Components: Nozzles

Convergent Nozzles

- Flow accelerates to throat
- Depending on nozzle pressure ratio, nozzle is either choked or unchoked

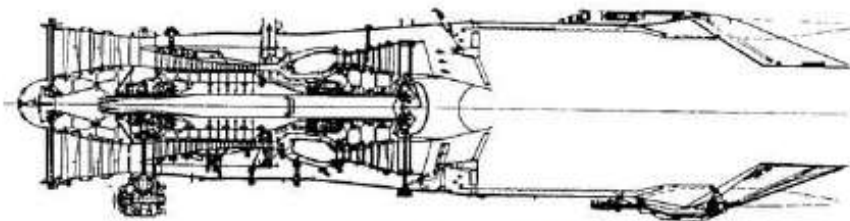
Convergent Divergent (Con-Di) Nozzles

- Initially converges to throat and then diverges
- Flow accelerates to $M = 1$ at throat and then continues to be accelerated in divergent section for full expansion

Which one is better? Trade off between

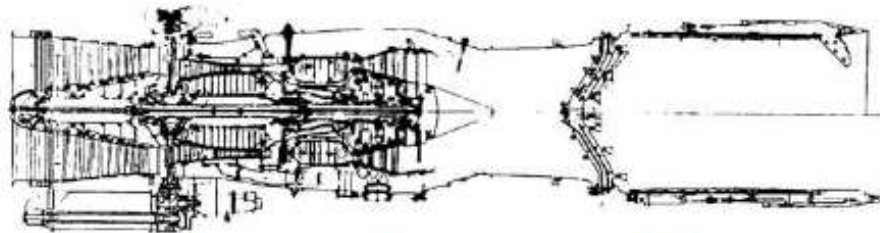
- Pressure thrust augmentation
- Weight and length penalty
- Ease of installation of variable area mechanism and thrust reversers

Afterburner Components: Nozzles



PRATT & WHITNEY F 100 (F-15)

- Dry thrust ≈ 79 kN
- Wet Thrust ≈ 129 kN
- Con-Di Nozzle



ROLLS-ROYCE SPEY Mk. 202 (PHANTOM)

- Dry thrust ≈ 54 kN
- Wet Thrust ≈ 91 kN
- Convergent Nozzle

Example: Cooling Requirements and Impact on Thrust

The lifecycle of an afterburner is limited by the excessive temperature in the casing. An analysis showed that in order to extend it, the cooling flow should be increased from 12% to 24% of the total flow.

1. Calculate the anticipated exhaust temperature and comment on the total reheat thrust reduction, assuming that the convergent nozzle is choked on both cases. The ambient temperature is 275 K and the gas generator exhaust temperature is 920 K. Combustion efficiency on the afterburner is 87%.

2. Discuss changes considered to recover the original thrust.

› Old Afterburner Configuration	› New Afterburner configuration
› $M1 = 88\%$	› $M1 = 76\%$
› $M2 = 12\%$	› $M2 = 24\%$
› $N_{cc} = 87\%$	› $N_{cc} = 87\%$

$M1$ = reheat flow

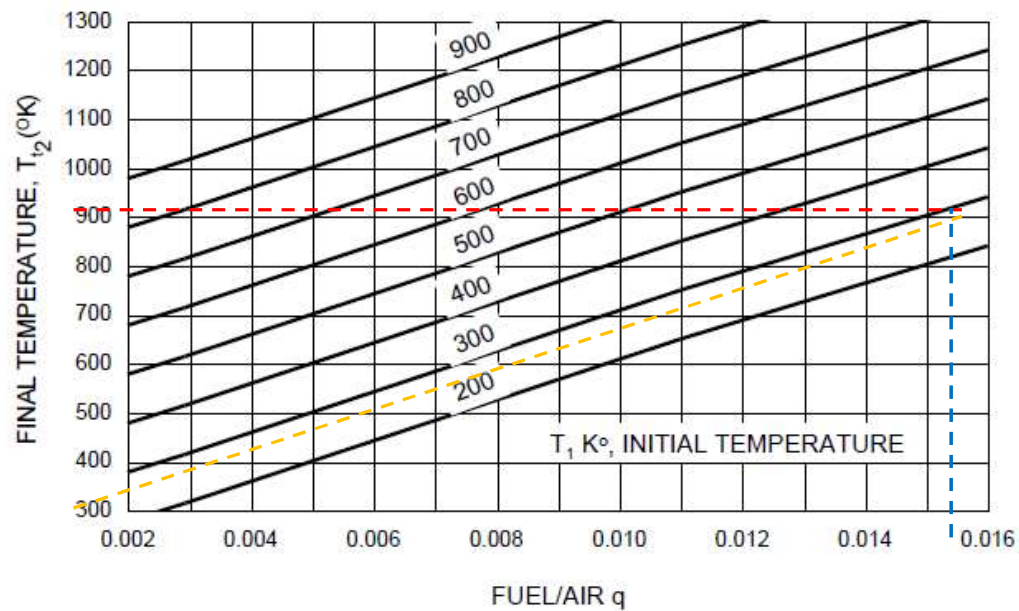
$M2$ = cooling flow

q_{OV} = overall fuel air ratio to increase from $T1$ to TR

q_E = fuel air ratio to increase from $T1$ to $T4$

N_{cc} is constant since the same combustor is being used.

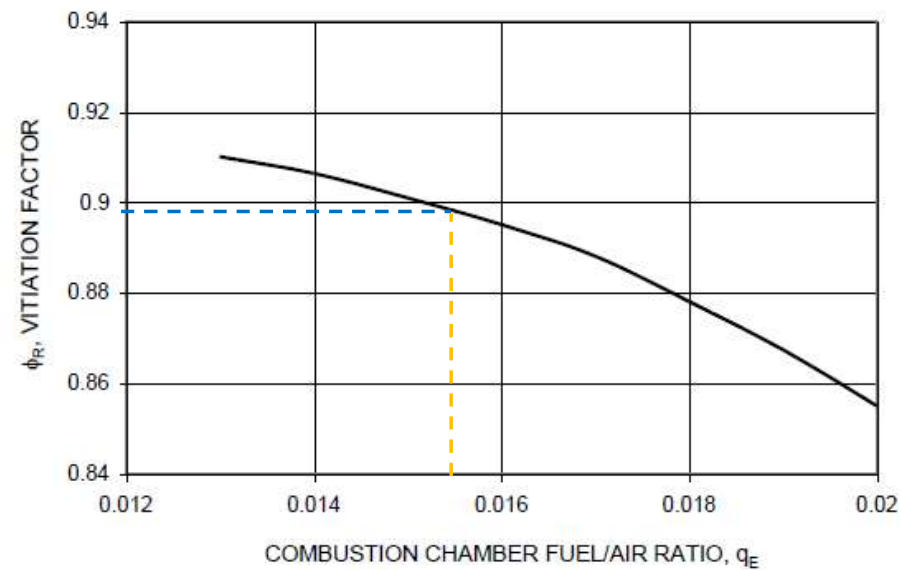
T_4 vs f_{GG}



Since initial temperature is 275 K and final gas generator is 920 K

From temperature rise characteristics we can obtain the fuel/air ratio (qE).

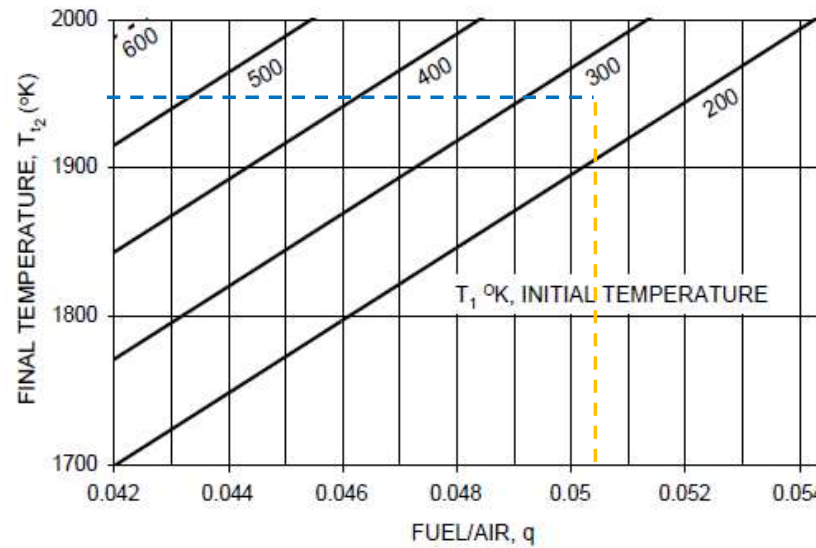
Vitiation factor vs f_{GG}



Using the fuel to air ratio previously obtained we can obtain the Vitiation factor.

Tr vs f_{ov}

$$\frac{\dot{m}_1}{\dot{m}_1 + \dot{m}_2} = \frac{f_{ov} - f_{GG}}{\eta_c(\phi_{AB} \cdot 0.0676 - f_{GG})}$$



From the jet pipe cooling flow we can estimate the fuel to air ratio of the overall engine, which will help us to determine the final reheat temperature in the afterburner.

Example: Cooling Requirements and Impact on Thrust

If we increase the cooling air flow, T_r tends to decrease due to the reduction of the air available for combustion. For the new design because we are using the same burner, T_1 , T_4 , η_{cc} , f_{gg} and ϕ remain constant. Repeating the above procedure we find that $T_{r_new} \approx 1825$ K.

The reheated thrust is calculated as follows:

$$F_N = m(V_E - V_0) + A_N(p_e - p_{atm})$$

assuming $A_N(p_e - p_{atm}) = K \cdot V_E$ where $(0 < K < 1)$

for a choked nozzle, $V_E \propto T^{0.5}$ y $A_N(p_e - p_{atm}) = \text{constant}$

Therefore comparing the thrust of both the old and new afterburner results as follows:

$$\frac{F_{old} - F_{new}}{F_{old}} = \frac{M(V_{E_{old}} - V_{E_{new}})}{MV_E + \text{constant}}$$

Example: Cooling Requirements and Impact on Thrust

So that:

$$\frac{F_{old} - F_{new}}{F_{old}} = \frac{\sqrt{1946} - \sqrt{1825}}{\sqrt{1946}(1 + K)}$$

Because K varies between 0 and 1, the maximum percentage that can vary the thrust is between 1.58% and 3.16%.

Conceptual Afterburners Design

We model an afterburner as a constant area duct with a series of bluff bodies in the stream with a known drag coefficient. Assuming the following inlet conditions for dry mode

- Inlet Mach number $M_i = 0.3$
- $\gamma_i = \gamma_e = 1.33$ and $C_{pi} = C_{pe}$
- And assuming a variation of Drag coefficient $C_d = [0.3 - 0.7]$

Determine

- Static pressure ratio variation to C_d
- Exit Mach number M_e
- Static Temperature ratio T_e/T_i
- Total pressure ratio P_e/P_i

Conceptual Afterburners Design

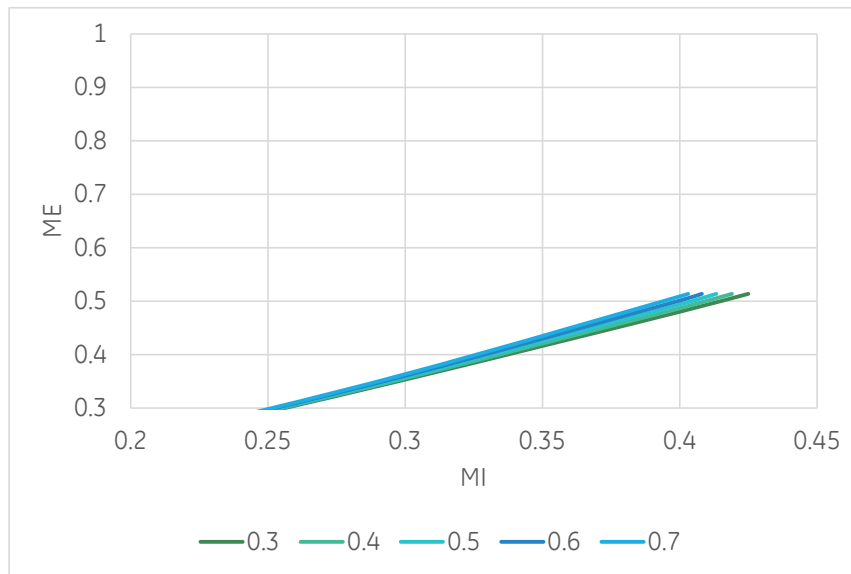
If the afterburner mentioned before operates in wet mode, and assuming the nondimensional heat release $q/C_p i T_i = 1.0$ and $\gamma_e = 1.25$, calculate

- Total pressure ratio in the wet mode
- Exit Mach number in the wet mode

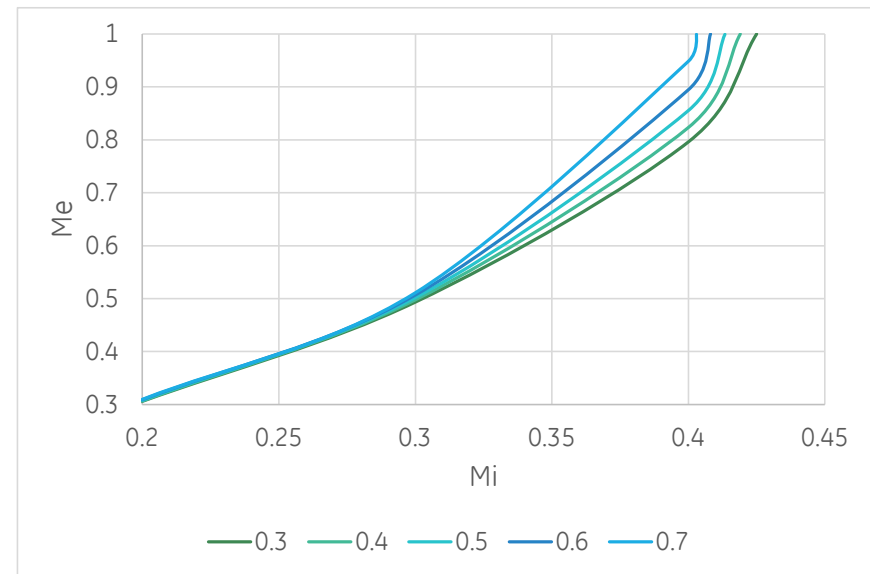
Discuss the differences for the results between the two operating points due to the changes in drag coefficient and conclude on the impact about the flameholders on the afterburners.

Conceptual Afterburners Design

EXIT MACH NUMBER AT CONSTANT HEAT INPUT (DRY MODE)

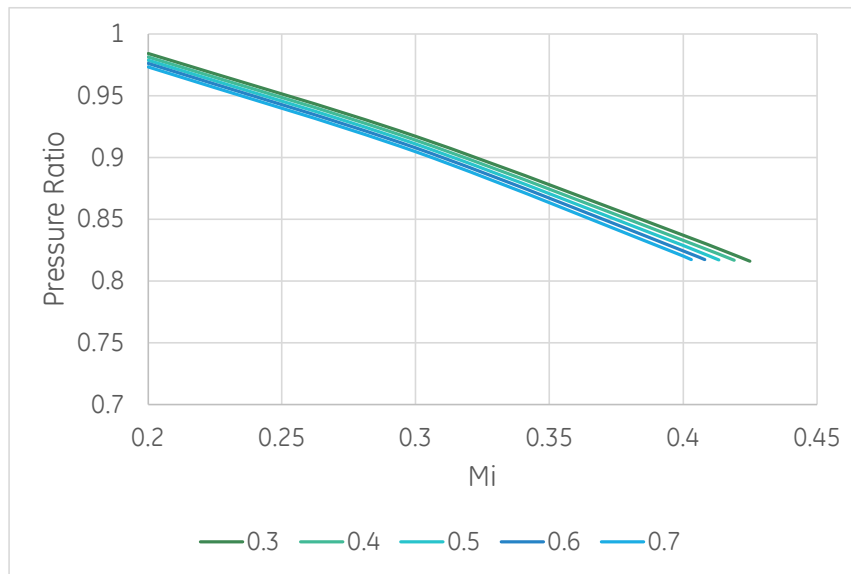


EXIT MACH NUMBER AT CONSTANT HEAT INPUT (WET MODE)



Conceptual Afterburners Design

PRESSURE RATIO AT CONSTANT HEAT INPUT
(DRY MODE)



PRESSURE RATIO AT CONSTANT HEAT INPUT
(WET MODE)

